

Organic Rice Network and Royal Rice Mill

Si Phum Sub-district · Tha Wang Pha District · Nan Province

From flood relief to organic rice — a learning journey

Background

Si Phum sub-district lies on the flat, irrigated floor of the Nan River valley in Tha Wang Pha district is one of the main rice-growing areas of Nan province. Farming families traditionally grew paddy rice for household consumption and sold any surplus, alongside fruit orchards (orange, litchi, longan), upland rice and cash crops such as tobacco. As elsewhere in Thailand, the move to commercial farming brought heavier use of agrochemicals: orchards gave high yields, but at a high cost to health and household budgets. Farmers who sold paddy depended on middlemen and private mills, and faced depressed prices and rising input costs.

At provincial level, Nan's farmers were caught in a vicious cycle of growing debts, overusing agrochemicals and deteriorating health. As farmland shifted to non-food cash crops, the province increasingly brought in food from outside, of unknown quality and unpredictable quantity; a real concern for food safety and food security.

The turning point for Si Phum came in 2006 (B.E. 2549), when flash floods and the overflowing Nan River struck northern Nan. Homes and farmland were badly damaged, and in Tha Wang Pha many farmers could not harvest their crops at all. The response to that disaster laid the foundations of both the community rice mill and, a few years later, the organic rice network.



Origins of the Mill and the Organic Network

A royal rice fund after the 2006 flood

After staff of the Private Secretary's Division of HRH Princess Maha Chakri Sirindhorn delivered royal relief supplies to flooded villages, they reported the scale of crop losses and proposed a **Royal Rice Fund**. The Princess granted money, milled rice and rice seed to start the fund, so that affected households could replant and the community would be better prepared for future disasters.

Mr. Son Khamin, then sub-district headman (kamnan) of Si Phum, played a key role in brokering this support and in persuading villagers to pool their own savings alongside it. Households that received royal seed later returned part of their harvest to the fund, so that other communities could benefit in turn.

From a rice fund to a community mill

Seeing how well the community managed the fund, and responding to villagers' wish for their own mill, the Princess granted a community rice mill. A CPR 1000 community-scale milling unit was donated through the Charoen Pokphand (CP) Group; the site, on high ground safe from floods, was secured with support from the Treasury Department and the Nan Provincial Administrative Organization; and the Si Phum Sub-district Administrative Organization initially paid the mill's staff. The **Royal Rice Mill** (โรงสีข้าวพระราชทาน) was formally opened by the Princess on 27 February 2009 (B.E. 2552), when she symbolically packed the first bags of rice.

The Nan Provincial Cooperative Office acted as mentor: it introduced the cooperative model and trained staff who had no prior background in milling, paddy purchase, accounting or marketing. The mill follows the Sufficiency Economy philosophy and the motto "Nan rice for Nan people". On a royal visit in 2010 (B.E. 2553), Mr. Son, then chairman of the mill, asked for help with irrigation; the Royal Irrigation Department went on to build the so-called "miracle canal" (เหมืองมหัศจรรย์), a weir and 8-km main canal worth over 30 million baht that eased water shortages for local farmers.





From chemical to organic rice

Mr. Chin Sutha "Po (Father) Chin" left school after grade 3 to farm with his parents, growing oranges, litchi, longan, hill rice and tobacco. He knew that heavy chemical use in the orchards was unhealthy and costly but carried on like his neighbours. A member of the mill since its founding, he holds ten shares. In 2007 the Provincial Land Development Office encouraged him to grow "safe" rice, which he did for two to three years.

> ***"I sell my organic rice to the mill and get a very good price — better than the market price."***
>
> Mr. Chin Sutha, member of the Royal Rice Mill since its founding.

Around 2010–11 (B.E. 2553–54), as he recalls, he attended a training at the mill led by Ajarn “Tik” (Dr Tansiphorn Na Nan), where he first heard about PGS Nan, and joined the Chulalongkorn University programme. His first organic season was a failure: not yet knowing how to control weeds, and unsure he could manage pests, he harvested nothing. Government training at the mill then taught him to control weeds through water management, and yields improved in the following years. By 2013 (B.E. 2556) he was confident enough in the network to convert a second plot, a mixed fruit orchard, to organic.

He also brought his older brother, Mr. Bunchuay Sutha, a former construction contractor whose main income comes from rubber plantations, into PGS Nan; Mr. Bunchuay farms only his paddy fields organically.

Organic Rice under PGS

The School of Agricultural Resources of Chulalongkorn University, through its Nan learning centre, joined provincial meetings on the farming crisis and proposed organic farming certified through a Participatory Guarantee System (PGS), a locally focused, peer-based guarantee built on trust, social

networks and knowledge exchange. Because local farmers knew little about organic farming, the School chose a participatory approach: from 2014, fourth-year undergraduates carried out their senior projects with rice farmers in Tha Wang Pha, documenting their practices, introducing organic methods and designing an organic rice business model based on PGS. Over two years the group was convinced to change its practices, and the idea spread to other districts.

The School also secured a market by partnering with the Lemon Farm Organic PGS network, run by the Health Society Company, which operates organic shops in and around Bangkok. Formal training began in March 2017; in December 2017 the "PGS organic agriculture Nan Group" was established with 35 households in six districts, including Tha Wang Pha, and in March 2018 it was certified by Lemon Farm Organic PGS.

Members of the PGS group guarantee each other's produce: they visit and audit one another's farms, plan production together and carry out quality control, backed by training on organic standards, farm management and certification. As members increasingly learned from each other through peer review and learning by doing, the university's role shifted from teacher to advisor and facilitator. The mill's hall hosts training on organic rice and on water management for weed control, and certified organic paddy is milled and sold by the mill, part of it to Lemon Farm.

TIMELINE SUMMARY

2006
B.E. 2549
Major flood in northern Nan; Royal Rice Fund set up with money, rice and seed.

2007
B.E. 2550
Mr. Chin starts growing "safe" rice with the Provincial Land Development Office.

2009
B.E. 2552
Royal Rice Mill opened by the Princess (27 Feb) and run as a member cooperative.

2010–11
B.E. 2553–54
Royal visit leads to the "miracle canal". Mr. Chin joins PGS Nan with Chulalongkorn.

2014–16
B.E. 2557–59
Chulalongkorn students work with Tha Wang Pha rice farmers; link with Lemon Farm.

2017
B.E. 2560
First PGS training; PGS organic agriculture Nan Group founded with 35 households.

2018
B.E. 2561
Certified by Lemon Farm Organic PGS (Mar); 46 households in 9 districts (Jun).

2019
B.E. 2562
"Krasib-Raks" brand and online sales as supply outgrows Lemon Farm demand.

Activities and Services

A one-stop service centre for rice farmers

In normal times the mill works as an all-round agricultural service centre for its members. Its business includes:

- **Paddy collection** from members and other farmers, at 0.50–1 baht/kg above the local market price depending on quality.
- **Milling and sale of milled rice** to over 30 regular public and private buyers.
- **By-products** (bran, broken rice, husk) and a **milling service** for households across Nan.
- **Farm supplies** (fertiliser, rice seed, general goods), **low-interest credit** for members.
- **Coffee processing** and the Phu Phayak coffee shop and souvenir shop at the mill's learning centre.

The mill processes around 6 tonnes of rice a day in a GMP-certified facility and sells safe and organic rice under the **"Khao Nan"** (ข้าวน่าน) brand, including Riceberry, red jasmine, white jasmine and purple glutinous rice (khao niao kam). Its roadside location makes it a popular stop for visitors exploring Tha Wang Pha.



Learning with young people

Since 2014 the mill has run youth training on rice and mill management. In May 2017, 200 secondary students from Tha Wang Pha Phitthayakhom School followed the whole chain from rice cultivation to the work of the mill, then applied the cooperative model to their own school.

Business Model and Networking

A cooperative enterprise

The Royal Rice Mill Group is owned and run by local residents following cooperative principles. Shares cost 10 baht each, with a minimum of ten on joining and no upper limit. Members are organised in 13 groups — one for each village of Si Phum (three of them in Ban Don Tan) and one for neighbouring sub-districts. In 2016–17 the mill had 1,196 members, a 15-member board drawn from village leaders, two auditors, a management team of four and three hired staff. Salaries and running costs were covered entirely by the mill's own income, without further government support, and it paid dividends every year since its founding without recording a loss.

Net profit is allocated each year by the general assembly: at least 10% to the reserve fund, at least 20% to the Royal Rice Fund's disaster reserve, at least 10% to a development fund, at least 5% each to member welfare and to public benefit, and the remainder to dividends on shares and to patronage refunds based on the paddy members sell. The coffee shop has become a significant earner: its revenue rose from 0.44 to 1.52 million baht between FY2014 and FY2015. In FY2016 the mill collected 359 tonnes of paddy (target 400) and sold 151 tonnes of milled rice (target 200), reflecting strict quality control on purchases.



Partnerships, buyers and support

As a royal project, the mill has enjoyed smooth relations with agencies at provincial, district and sub-district levels and with the private sector, bringing funding, knowledge and markets with little conflict. Key partners include the Office of HRH Princess Sirindhorn, the Nan Provincial Cooperative Office, Si Phum SAO, Nan PAO, the Royal Irrigation Department, the Provincial Land Development Office and the Rice Research Centre.

Current challenges

An organic market bottleneck. As the PGS group grew from 35 to 110 households in two years, supply quickly outran Lemon Farm's demand. Because Lemon Farm sells to small retail customers, it buys in small lots: organic rice built up in the mill's stock, which then could not take in new harvests, putting pressure on the mill. Some members have stepped back. Then, the organic rice market was hit hard during and after COVID as consumers could not afford high price premiums. Many members withdrew because organic volumes did not match purchases and switched to "safe standard" coloured rice with a wider market.

Ageing farmers and succession. Mrs. Jurirat Intrawang "Mother Jui" came back from 20 years in Bangkok when her father fell ill; she grows organic rice but her children are not interested in farming, so she will continue "as long as her body allows". The mill faces the same question: young people leave for the city, capable villagers decline board seats, members tend to re-elect the same village leaders, and the sense of ownership is weaker than in mills built solely from villagers' own savings.

Reccuring disasters have also dramatically constrained rice production in the area. Mr. Chin and his neighbours were hit by heavy floods and did not get any rice harvest three years in a row in 2024, 2025 and 2026 which dramatically reduced the volumes of organic rice treated by the Royal Mill, only 600 kg in 2025, which put the initial business model at risk.

